

JERÓNIMO GARCÍA-MEJÍA

PERSONAL DATA

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RESEARCH INTERESTS: Geometric Group Theory and Geometric Topology.

ACADEMIC POSITIONS

OCT 2025 – PRESENT UNIVERSITY OF WARWICK
Research Fellow
Mathematical Institute
Postdoctoral Mentor: Dr. Ric Wade
DEC 2024 – OCT 2025 UNIVERSITY OF OXFORD
Postdoctoral Research Associate
Mathematical Institute
Postdoctoral Mentor: Dr. Ric Wade

EDUCATION

SEP 2024 KARLSRUHE INSTITUTE OF TECHNOLOGY
PhD in Mathematics
Thesis: *Dehn functions of groups with filiform subgroups*
Advisor: Jun.-Prof. Claudio Llosa Isenrich
AUG 2020 UNIVERSITY OF BONN
M.Sc. in Mathematics
Thesis: *Group actions on hyperbolic spaces*
Advisor: Prof. Ursula Hamenstädt
APR 2018 UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO
B.Sc. in Mathematics
Thesis: *Construction of classifying spaces* (in Spanish)
Advisor: Prof. Carlos Prieto

SCHOLARSHIPS

2018–2020 ROSA LUXEMBURG STIFTUNG (Rosa Luxemburg Foundation)
Scholarship

PUBLICATIONS

- Yu-Chang Chan, Jerónimo García-Mejía, and Matteo Migliorini, *Complete classification of the Dehn functions of Bestvina–Brady groups*.
Geometric and Functional Analysis (2026).
- Jerónimo García-Mejía, Claudio Llosa Isenrich and Gabriel Pallier *On Dehn functions of central products of nilpotent groups*.
Transactions of the American Mathematical Society (2026).
- Jerónimo García-Mejía and Antoine Goldsborough, *Random Dehn function of groups*.
Journal of Topology and Analysis (2026).

OTHER WORK CONTAINING ORIGINAL RESEARCH

- *Dehn functions of groups with filiform subgroups.*
PhD thesis, Karlsruhe Institute of Technology.
- *Group actions on hyperbolic spaces.*
Master thesis, University of Bonn. Unpublished.

INVITED TALKS

- OCT 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, **Conference: LiT IV Landing in Topology**, University of Pisa (scheduled).
- SEP 2026 *TBD*, **Conference: Automorphisms groups in Bonn**, University of Bonn (scheduled).
- JUN/JUL 2026 *La geometría del problema de la palabra* **Summer School: Escuela de Teoría Geométrica de Grupos**, Instituto de Matemáticas Oaxaca, México.
- MAY 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, **Geometry and Topology Seminar**, University of Glasgow.
- APR 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, **Oberseminar Geometric Group Theory**, University of Bonn.
- JAN 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, **Topology Seminar**, University of Oxford.
- JAN 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, **Geometry and Topology Seminar**, University of Warwick.
- NOV 2025 *Complete classification of the Dehn functions of Bestvina–Brady groups*, **Algebra Seminar**, University of Bristol.
- MAY 2025 *Dehn functions of Bestvina–Brady groups*, **Geometric Group Theory Seminar**, University of Cambridge.
- JULY 2024 *Dehn functions of nilpotent groups*, **Geometric Group Theory Seminar**, University of Oxford.
- APR 2024 *Dehn functions of nilpotent groups*, **Research seminar Model theory and groups**, Universität Münster.
- MAR 2024 *Funciones de Dehn de grupos nilpotentes*, **Long-distance seminar in Geometric Group Theory in Mexico**.
- JAN 2024 *Dehn functions of central products of nilpotent groups*, **Geometry and Dynamics Seminar**, Université de Lille.
- OCT 2022 *The Geometry of the Word Problem*, **Baby Geometri Seminar**, University of Pisa - Scuola Normale Superiore.
- MAY 2020 *Group actions on hyperbolic spaces*, **Junior Geometry Seminar**, Heidelberg University.

CONTRIBUTED TALKS

- MAY 2026 *Lightning Talk*, IMBM Istanbul Topology Days:: A Conference on Developments in Contact Topology, Mapping Class Groups, and Beyond, Boğaziçi University, Istanbul, Turkey.
- MAY 2023 *Lightning Talk*, Conference Geometry of subgroups, Thematic program on Geometric Group Theory, Centre de Recherches Mathématiques, Montréal.
- FEB 2023 *Lightning Talk*, Young Geometric Group Theory XI, Münster.
- FEB 2021 *Group actions, hyperbolic spaces and small cancellation*, AG Seminar Metric Geometry, Karlsruhe Institute of Technology, Karlsruhe.

CONFERENCES AND WORKSHOPS ATTENDANCE

- MAY 2026 *IMBM Istanbul Topology Days: A Conference on Developments in Contact Topology, Mapping Class Groups, and Beyond*, Boğaziçi University, Istanbul, Turkey.

MAR 2026	<i>Topological and Homological Methods in Group Theory 2026</i> , Bielefeld University, Bielefeld, Germany.
MAR 2026	<i>Journées Trans-Couesnon de théorie des groupes</i> , University of Caen, Caen, France.
SEP 2025	<i>Actions on graphs and metric spaces</i> part of the program <i>Operators, Graphs, Groups</i> , Isaac Newton Institute for Mathematical Sciences, Cambridge, England.
JUL 2025	<i>Non-positive curvature and applications</i> , Isaac Newton Institute for Mathematical Sciences, Cambridge, United Kingdom.
JUN 2025	<i>The William Rowan Hamilton Geometry and Topology Workshop: Celebrating Martin Bridson's 60th Birthday</i> , The Hamilton Mathematics Institute, Trinity College Dublin, Dublin, Ireland.
MAY 2025	<i>Groups and Geometry in South England</i> , University of Oxford, Oxford, England.
AUG 2024	<i>Workshop on Profinite Rigidity</i> , Karlsruhe Institute of Technology, Karlsruhe, Germany.
APR 2024	<i>Young Geometric Group Theory XII</i> , University of Bristol, Bristol, England.
MAR 2024	<i>Topological and Homological Methods in Group Theory 2024</i> , Bielefeld University, Bielefeld, Germany.
JUN 2023	<i>Cube complexes and combinatorial group theory</i> part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Montréal, Canada.
JUN 2023	<i>Groups around 3-manifolds</i> part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Montréal, Canada.
MAY 2023	<i>Geometry of subgroups</i> part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Montréal, Canada.
FEB 2023	<i>Young Geometric Group Theory XI</i> , Münster University, Münster, Germany.
JUL 2022	<i>Metric Geometry and Geometric Analysis</i> , MSRI-Oxford Summer Graduate School, Oxford University, Oxford, England.
JUN 2022	<i>Hyperbolic groups and their generalisations</i> , Institut Henri Poincaré, Paris, France.
MAR 2022	<i>Topological and Homological Methods in Group Theory 2022</i> , Bielefeld University, Bielefeld, Germany.
FEB 2022	<i>Workshop on finiteness properties</i> , Karlsruhe Institute of Technology, Karlsruhe, Germany.

EXTENDED VISITS

OCT 2025 – JAN 2026	UNIVERSITY OF OXFORD, MATHEMATICAL INSTITUTE Academic visitor
SEP 2025	ISAAC NEWTON INSTITUTE Semester program <i>Operators, Graphs, Groups</i>
MAY – JUN 2023	CENTRE DE RECHERCHES MATHÉMATIQUES Thematic program on <i>Geometric group theory</i>

TEACHING EXPERIENCE

My teaching experience as *Teacher's Assistant* covers preparing exercise-sheets, presenting solutions in front of the class, and grading. All of this in Spanish, English, and German.

WINTER 2021/22	<i>Elementare Geometrie (Elementary Geometry)</i> , Bachelor course in German at the Karlsruhe Institute of Technology.
SUMMER 2021	<i>Geometric Group Theory</i> , Master course in English at the Karlsruhe Institute of Technology.
WINTER 2020/21	<i>Proseminar Geometrie: Knotentheorie (Knot theory)</i> , Bachelor Seminar in German at the Karlsruhe Institute of Technology.
WINTER 2017/18	<i>Analytical Geometry I</i> and <i>Topology I</i> , Bachelor courses in Spanish at Universidad Nacional Autónoma de México.

SERVICES

CURRENT	I am supervising an undergrad summer research project in Mathematics.
JUL - AUG 2025	I supervised of a group of three students as part of a summer research school for women in Mathematics.
AUG 2024	I co-organised of the workshop Profinite Rigidity, joint with the University of Münster, Regensburg University, and the Karlsruhe Institute of Technology, Karlsruhe, Germany.
APR 2022	I co-organised of the RTG "Asymptotic Invariants and Limits of Groups and Spaces" Seminar: Group (co)homology joint with the Karlsruhe Institute of Technology and Heidelberg University, Germany.
OCT 2021 – SEP 2022	I was a speaker of the RTG "Asymptotic Invariants and Limits of Groups and Spaces".